BraTS 2024 Synthesis Project - Weekly Report

Group Name: Vision disruptors
Project: Synthesizing Missing MRI Modalities for Brain Tumor Segmentation
Report Period: Week 3
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Project Overview

This week, we have spent most of our time developing the current architecture and a roadmap for achieving it. In pursuit of this end, we hope to develop a deep learning architecture capable of producing high-fidelity synthesized MRI images without anatomical errors or loss of tumor pathology. The training model will be trained on the BraTS 2024 dataset, which includes brain MRI scans in multiple modalities and tumor segmentation annotations.

Proposed Architecture

To address the issue of producing missing MRI modalities, we introduce a new architecture, the Tumor-Aware Multi-Scale Wavelet Diffusion Model (TAMS-WDM). This structure combines multi-scale wavelet processing, cross-modal attention, cross-modal contrastive learning, and diffusion-based image generation. The model effectively learns to understand relationships between MRI modalities and directs attention to the tumor regions important for medical analysis, thanks to its design.

The architecture is built around four primary steps, each with a specific responsibility in the synthesis pipeline.

Stage 1:

Performs a three-dimensional discrete wavelet transform (DWT) on the input MRI modalities. The wavelet decomposition technique decomposes the image into various spatial-frequency components that can be manipulated at different levels of resolution. This model can store minute details as well as large body parts, since it requires little memory to run. The decomposition process is carried out at three levels: the first, which

operates at the fine scale, learns to identify fine details and differentiate tumors from surrounding tissue.

Stage 2:

which functions at the medium-scale level 2 is trained to learn patterns of human anatomy between small features and overall anatomy features. The third level, coarse scale level 3, is trained to recognize the entire brain structure. The following output from the feature representations, as discussed, has a dimension of 120×120×78, representing every minute detail. The second level of detail presents an output of 8, with dimensions of 60 x 60 x 39, indicating all medium-sized structures. The third level of detail indicates an output of 8, with 30×30×20 dimensions, that presents all major body structures. The wavelet decomposition algorithm enables users to achieve an 8x reduction in memory usage while retaining all data from the original image, making it well-suited for processing large volumes of 3D medical data. The model learns the relationship between various MRI modalities using cross-modal attention. The tissue properties that are depicted by each modality render it necessary to apply appropriate techniques to synthesize their combinations. The model employs multi-head cross-attention (8 heads and 256 channels on the features), enabling the multi-modal features to attend to relevant information across different modalities. The network employs this capability to identify more scanning patterns that exist between the two scanning techniques.

Stage 3: Diffusion Generation that is tumor-aware.

A 3D U-Net diffusion model performs the core generation operations with 1000 denoising timesteps. The diffusion models produce images by progressively removing noise from an initial noisy image until a final clean image is obtained.

The system relies on a tumor-aware attention system that enables effective tumor reconstruction. The system automatically identifies tumor locations, paying 10x more attention to the identified areas, which makes the network focus on critical medical areas.

The diffusion model employs sinuoidal embeddings as time-step conditioning, allowing the network to moderate its advancement during the denoising procedure throughout training.

Stage 4: Multiscale Fusion/Reconstruction.

A learned fusion network is then used at the final step to combine outputs from different wavelet scales. Features from fine, medium, and coarse scales are used to produce a full-resolution image within the network.

The fusion module progressively downsamples feature channels by 24, then by 16, down to 8, and finally performs an inverse wavelet transform (IDWT). Reconstruction of the synthesized MRI scan is done to recreate it at its original resolution:

240 x 240 x 155

The last output symbolizes the created missing modality T1ce composed of the existing MRI scans.

Upcoming week's task

The project requires us to build core infrastructure elements, including PyTorch 2.1.0 and MONAI 1.3.0, along with all necessary dependencies for CUDA-enabled GPU support. The complete data-loading system for the BraTS 2024 dataset will handle preprocessing tasks and data augmentation, including percentile clipping, z-score normalization, random flips and rotations, and intensity shifts. The 3D wavelet transform module will be implemented and tested using perfect reconstruction tests, which require an error rate below 1e-6 for its single-scale and multi-scale decomposition functions. The 3D U-Net architecture will develop a model containing approximately 80 million parameters, with 4 encoder levels, a bottleneck, and 4 decoder levels that use skip connections. The training infrastructure will include a trainer class that implements gradient clipping, learning rate scheduling, and TensorBoard logging and checkpoint saving, along with SSIM, PSNR, MSE, and MAE evaluation metrics, and visualization tools for comparing real and synthetic images.